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Characterization of the Ductile and Brittle Behavior of Shale Through Plastic Strain, Mineralogy and Gas Flow

Francis Elisabeth and Pathi Venkat, Saudi Aramco

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Summary

Brittleness of reservoir rocks is one of the key rock properties essential for placement of horizontal wells and hydraulic stimulation to propagate fractures and create complex fracture network. This study presents a comprehensive method for characterizing the shale brittleness by analyzing the magnitude of plastic strain from rock mechanics tests, mineral content and benchmarked against gas flow of wells. Laboratory rock mechanics tests, including multi-stage and triaxial tests, were conducted to measure the plastic deformation characteristics of shale samples. These results were integrated with mineralogical data obtained from X-ray diffraction (XRD) and total organic carbon (TOC) analysis to quantify the magnitude of a fit for purpose brittleness. The ductile to brittleness transition was normalized and validated against gas flow from hydraulic fracturing in wells placed in different clay content intervals. Finally, this validated workflow was applied to a geologically analogous shale in a separate basin, enabling the identification of brittle intervals for optimal lateral well placement. The correlation between plastic deformation, mineralogical composition, organic content and brittleness were successfully calibrated with hydraulic fracturing performance, highlighting the critical impact of quartz, clay, and organic content on shale behavior. Quartz content was positively correlated with higher brittleness, while increased clay and organic matter content contributed to more ductility. The workflow enables reliable identification and quantification of brittle zones for hydraulic fracturing and lateral well placement, minimizing the risk of targeting ductile lithologies. Application of this methodology to a similar shale in another basin demonstrates its effectiveness as a benchmarking tool for unconventional reservoir characterization. This integrated approach enhances decision-making for efficient and sustainable hydrocarbon recovery. This study introduces a unique methodology that directly quantifies brittleness by linking plastic deformation measurements with mineralogical composition and validates the approach with hydraulic fracturing outcomes. Unlike traditional brittleness indices that rely solely on elastic properties or mineralogy without direct quantification, this approach incorporates the role of ductile deformation mechanisms, providing a more robust characterization of shale behavior.

Introduction

The characterization of rock brittleness has been a topic of interest in the oil and gas industry, particularly in shale formations, since the unconventional revolution of the late 2000s (Sone and Zoback 2013). When

hydraulically stimulated, a brittle rock will shatter, creating multiple pathways for hydrocarbon flow if propped, whereas a ductile rock may exhibit no borehole breakdown or a loss of permeability due to its plastic nature. Due to the lack of direct measurements and the availability of geophysical logs, various "brittleness index" have been developed using elastic properties, strength, and mineralogy among other rock properties (Meng et al. 2015; Zhang et al. 2016). However, these methods may not necessarily quantify the true brittleness and ductility of the rock. This study attempts to quantify the true brittleness and ductility of our shale formation by analyzing the actual plastic deformation of rock samples during rock mechanics testing. Then, by correlating the measured plastic deformation with elastic properties, strength and mineralogy, we established a calibrated plastic strain profile (Inverse of brittleness) that correspond to the relative magnitude of brittleness for the entire vertical extension of our shale. Our results show a correlation between brittleness and well performance, with wells placed in areas of high brittleness exhibiting higher performance. This approach provides a more direct and robust characterization of shale brittleness, enabling better decision-making for hydraulic fracturing and unconventional reservoir development. This highlights the importance of accurate brittleness characterization in optimizing hydraulic fracturing operations and maximizing recovery from unconventional reservoirs.

Theory and Method

Brittleness and ductility are fundamental behavior of rocks that describes their ability to deform and break under stress. Brittle rocks behave elastically until they reach a breaking point, at which they fail abruptly and release a significant amount of energy (Wang and Park 2001; Zhang and Fu 2008; Meng et al. 2015; Zhao et al. 2015). This sudden failure is typically characterized by a sharp, vertical drop on a stress-strain graph. When subjected to stress, brittle rocks are prone to sudden fracturing or shattering (Elisabeth and Elena 2023). In contrast, ductile rocks exhibit a more gradual response to stress, deforming plastically and absorbing energy over time. This behavior is illustrated by a curved stress-strain curve, which shows a steady increase in deformation. When subjected to stress, ductile rocks tend to deform slowly, bending, folding, or flowing in a way that dissipates energy gradually. A typical illustrative example would be the one of a toothpaste to describe the ductile behavior, deforming smoothly under stress without fracturing. This illustrates high strain tolerance before failure. and a shattered glass for the brittle behavior. The material fractures suddenly with minimal plastic deformation, indicative of brittle failure under stress, Figure 1(left) and Figure 1(right).

Brittleness and ductility are crucial parameters in the oil and gas industry (Elisabeth and Elena 2023), particularly in unconventional reservoirs, as they significantly impact the effectiveness of hydraulic fracturing operations (Jin et al. 2014; Alghamdi and Elisabeth 2023). Figure 2 illustrates the structural differences and their implications for fluid flow in unconventional reservoirs, brittleness plays a key role in determining the outcome of hydraulic fracturing, with higher brittleness increasing the likelihood of creating complex fracture networks during stimulation. In tight sandstone formations with limited natural fractures, brittleness is likely to influence the length of the hydraulic fracture, with fractures placed in the most brittle intervals expected to exhibit longer fracture lengths. In contrast, in tight shale formations with pre-existing natural fractures, brittleness contributes to the overall complexity of the fracture network, leading to a more intricate and potentially productive fracture system [e.g., Figure 1(left)]. In ductile rocks, well performance impairment can occur due to the loss of permeability resulting from proppant embedment or the gradual closure of fractures over time caused by plastic deformation (Aramahi et al. 2012). This deformation can lead to a reduction in fracture aperture, ultimately compromising the long-term productivity of the well.

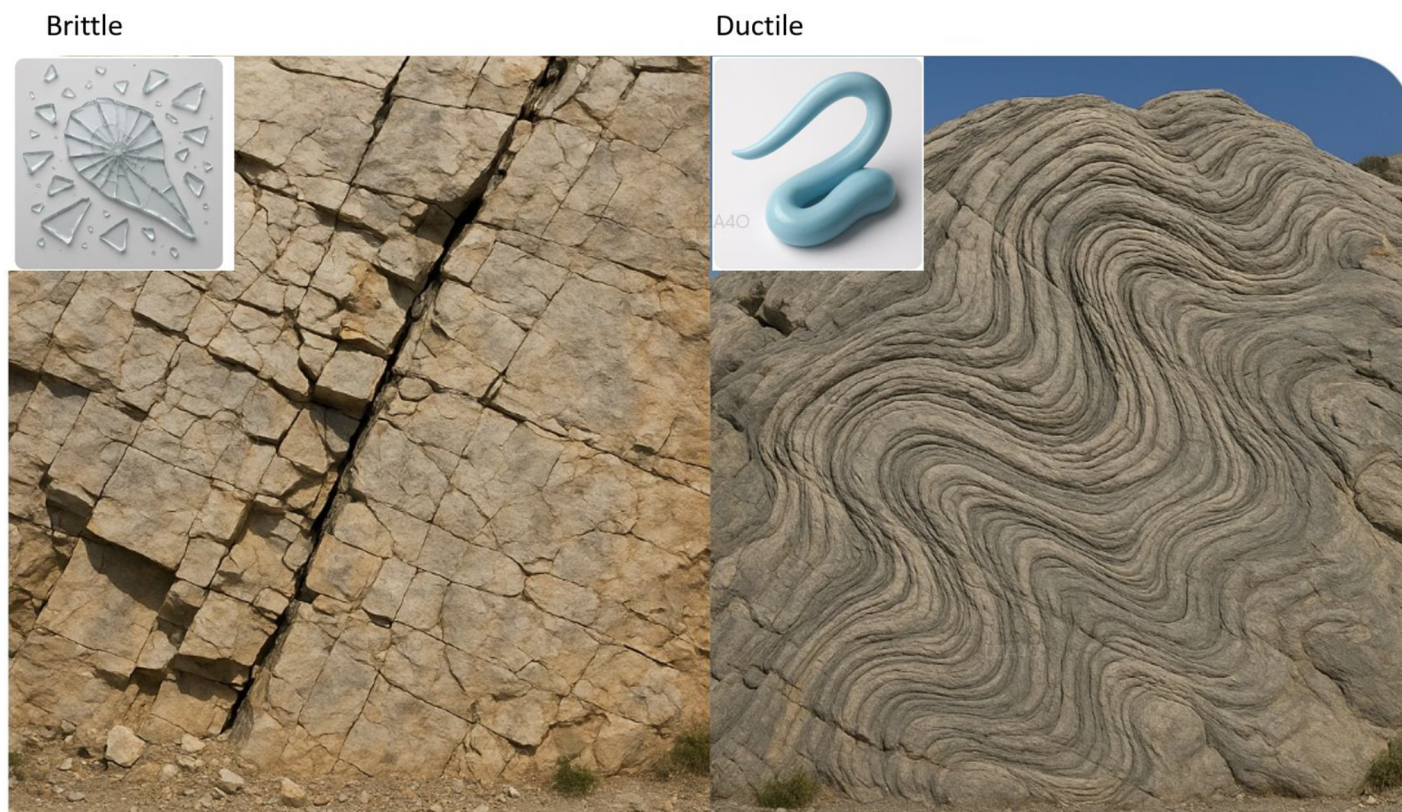


Figure 1—Illustration of a brittle rock (left) and deformed ductile rock (right).

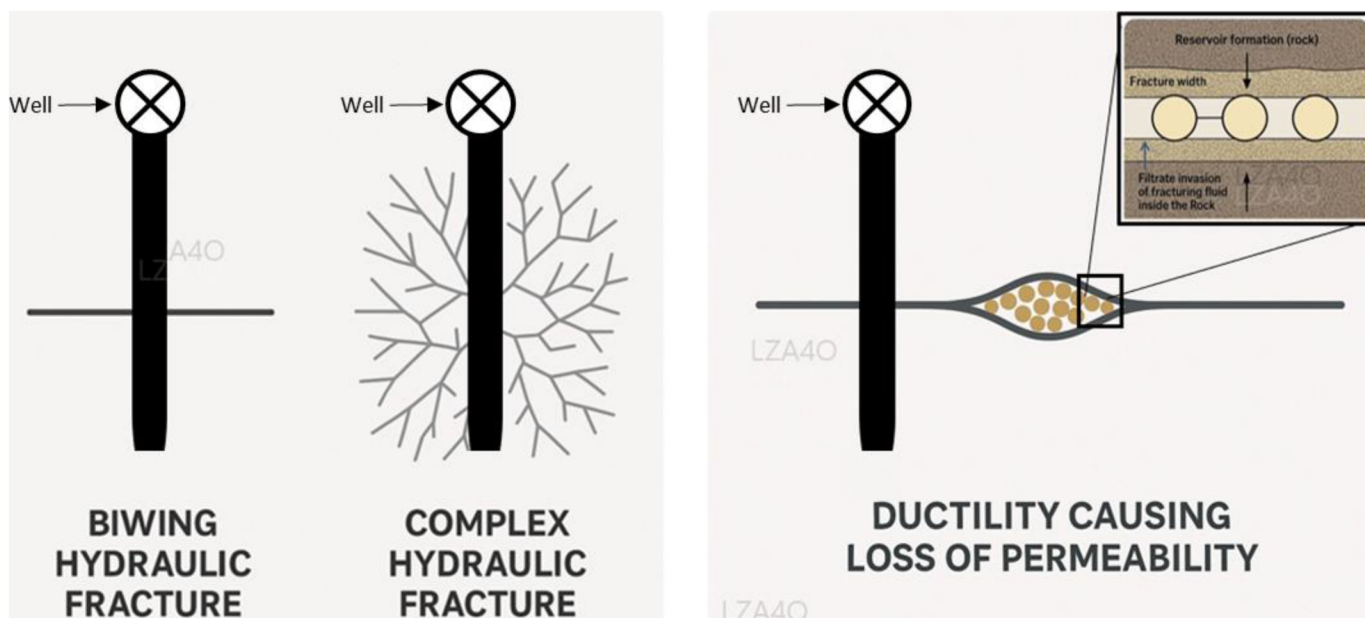


Figure 2—Schematic illustration of a bi-wing and complex hydraulic fracture networks (left) and embedment of proppant grains into softer formation rock under compressive stress, decreasing fracture conductivity (right) – (modified from [Katende et al. 2021](#)).

Several methods have been developed to quantify brittleness, incorporating a range of properties including elastic characteristics, rock strength, mineralogy, and other parameters. Elastic properties, such as Young's modulus (E), Poisson's ratio (ν), and shear modulus (G), are commonly used to estimate a brittleness index, as they provide valuable insights into a rock's ability to deform elastically. Additionally, rock strength

parameters, including UCS and tensile strength, can also be employed to estimate a brittleness index, as they reflect a rock's resistance to failure (Bai 2016). Other researchers have utilized the mineralogical composition of the rock, particularly the presence and abundance of quartz, clay, and organic matter, as key parameters to estimate brittleness index. These mineralogical components can significantly influence a rock's mechanical behavior, with quartz-rich rocks tend to be more brittle, while clay-rich rocks tend to be more ductile (Rickman et al. 2008).

In this study the brittleness is quantify through rock mechanics testing, which provides a more direct and accurate measurement of a rock's brittleness. Triaxial testing, a technique that involves subjecting rock samples to varying stress conditions, from the elastic domain to the plastic domain, and ultimately to failure, can be used to measure brittleness directly. This approach allows for a comprehensive understanding of a rock's mechanical behavior, including its elastic and plastic deformation characteristics. In rock mechanics testing, brittleness is inversely proportional to the magnitude of plasticity that a rock exhibits from yielding until failure. Specifically, a shorter region of deformation between the yielding point and the failure point indicates higher brittleness, as it suggests that the rock is more prone to sudden failure rather than gradual deformation. Conversely, a longer region of deformation between the yielding point and the failure point indicates lower brittleness, as it suggests that the rock is more capable of absorbing energy and deforming plastically before failing. The concept of the relationship of brittleness and ductility in a stress-strain graph is illustrated in Figure 3. The brittle curve shows a sudden drop after peak stress, while the ductile curve demonstrates extended deformation before failure.

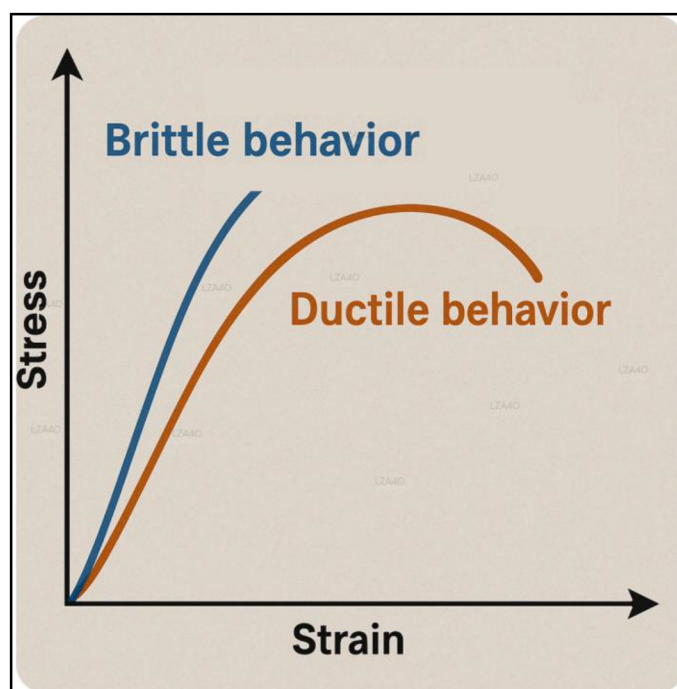


Figure 3—Stress-strain curve comparison for brittle and ductile materials.

Rock samples were carefully selected with multi-stage triaxial testing tested at downhole in-situ conditions to ensure that the results would be representative. The methodology involved interpreting key rock mechanics properties, including E , ν , friction angle (Fang), tensile strength (Tstr) and UCS, for each sample. These properties provide valuable insights into the mechanical behavior of the rocks and are essential for correlating with brittleness. Furthermore, volumetric strains were estimated as the sum of the radial and axial strains, which allowed us to quantify the overall deformation of the rocks. Plastic strains

were calculated as the difference between the volumetric strain at failure and the volumetric strain at the start of yielding, normalized against the volumetric strain at yielding.

$$\text{Plastic strain} = \frac{\text{volumetric strain at failure} - \text{volumetric strain at onset of yielding}}{\text{volumetric strain at onset of yielding}} \quad (1)$$

This approach enabled us to assess the plasticity of each sample and estimate its brittleness.

Figure 4 presents a comparison of the estimated plastic strain (inverse of brittleness) in four rock samples, along with images of the samples after triaxial testing. The results show distinct differences in the plasticity and brittleness of each sample. For example, Sample AQ11 exhibits the least plasticity, with multiple planes of failure consistent with a brittle behavior. In contrast, Sample AQ12 displays more plasticity and a minimum number of planes of failure. Notably, both samples are considered brittle, but with varying intensities of brittleness. This highlights the importance of quantifying brittleness, as even within a single rock type, there can be significant variations in mechanical behavior. Figure 4 clearly demonstrates the dependency of plasticity on the brittle behavior of rock, underscoring the need for a comprehensive understanding of rock mechanics properties to accurately predict and characterize reservoir behavior.

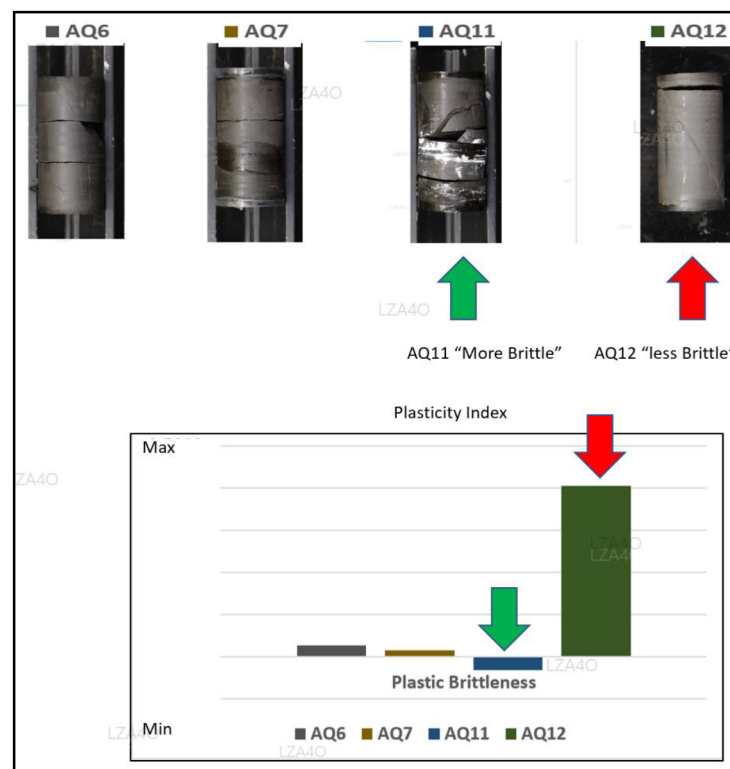


Figure 4—Plastic strain (inverse of brittleness) vs. sample failure - Top: Images of different rocks at failure. Below: A comparison of the estimated plastic strain in the four rock samples above.

The comparison of the samples also reveals that the plastic strain can be used as a reliable indicator of a rock's propensity for brittle failure. The plastic strains provide a quantitative measure of a rock's ability to deform plastically before failing, which is critical for predicting the likelihood of fracture initiation and propagation. Analyzing the brittleness with the lenses of plasticity in conjunction with other rock mechanics properties, gives a more complete understanding of the mechanical behavior of the rocks and make more informed decisions about reservoir development and production strategies.

A correlation of the estimated plastic strain with various elastic properties and rock strength parameters helped gain a comprehensive understanding of brittleness along the entire interval of our shale in each of our wells. This approach allowed us to identify the most significant relationships between these properties

and brittleness. Notably, we found that the best correlation existed between the plastic strain with E and also Rickman's² B19 equation for brittleness index that combine E and ν (Rickman et al. 2008). The Rickman's B19 equation is a widely used equation that relates brittleness index with the rock's elastic properties. The equation is based on the idea that brittleness is a function of the rock's ability to deform elastically and the stiffer the rock the more brittle. E and ν are normalized based on their minimum and maximum value. High value of B19 is commonly associated with brittle rock and low value of B19 as ductile or at least, least favorable for the landing point.

$$B_{19} = \frac{E_n + \nu_n}{2} \quad (2)$$

$$E_n = \frac{E - E_{min}}{E_{max} - E_{min}} \quad (3)$$

$$\nu_n = \frac{\nu_{max} - \nu}{\nu_{max} - \nu_{min}} \quad (4)$$

Figure 5 demonstrates a clear and consistent relationship between E and plasticity.

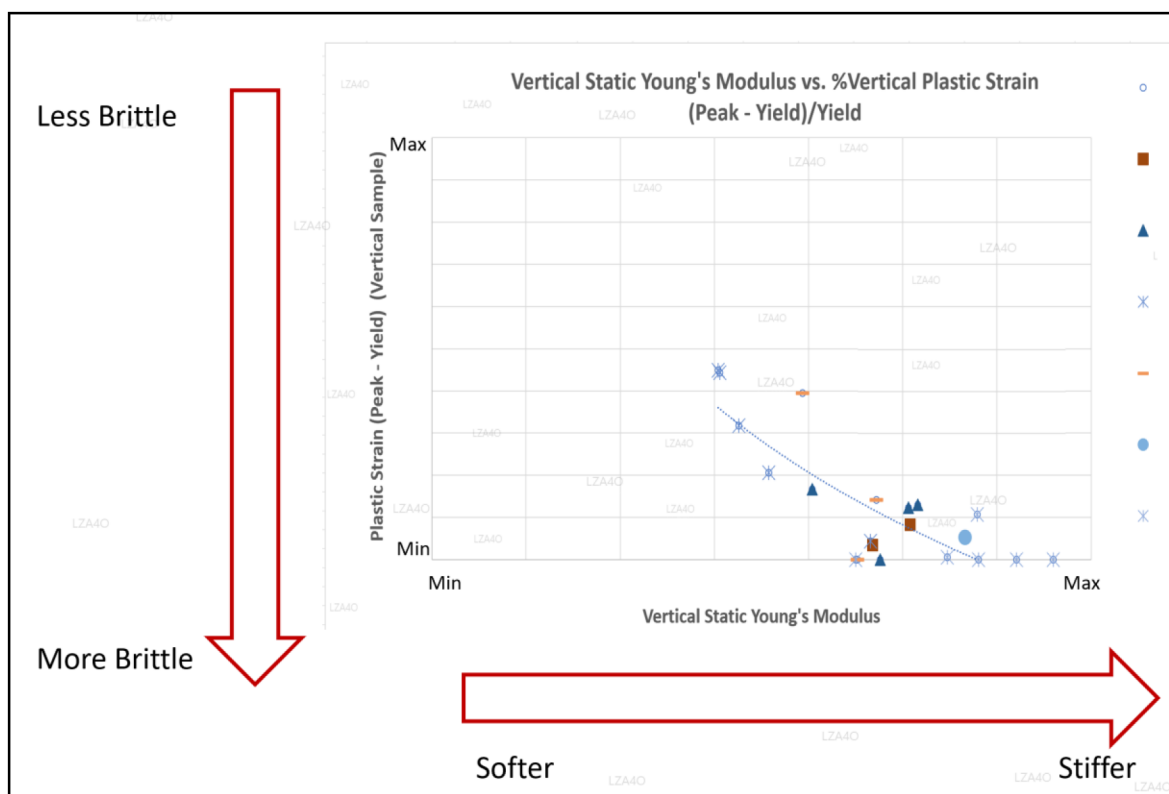
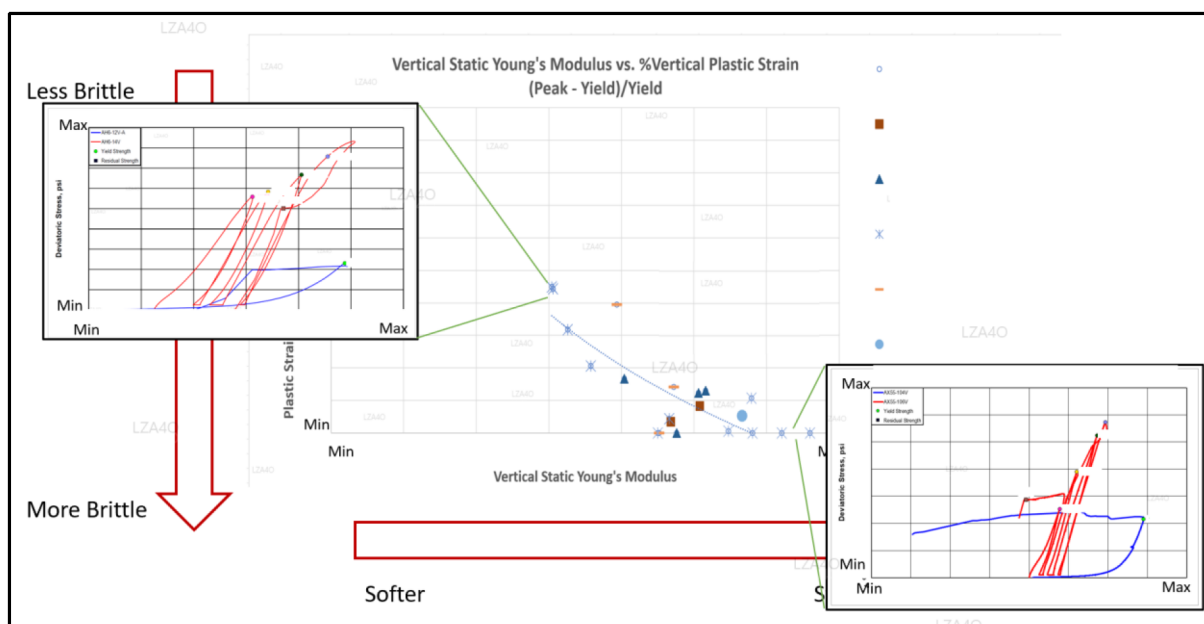


Figure 5— E vs plastic strain (brittleness).

Figure 6 is showing the same relationship of plasticity with E with the stress-strain graph for two wells with extreme values of brittleness, one representing the least brittle sample (high plasticity) and the other representing a more brittle sample (zero plasticity). As expected, the stress-strain graph shows a significantly higher plastic strain in the least brittle sample, indicating a greater ability to deform plastically and absorb energy before failing, whereas the more brittle sample exhibits no plastic strain, characteristic of a more sudden and catastrophic failure.



There is a good correlation between B19 and the plasticity of our samples, [Figure 7](#). A more comprehensive and accurate profile of brittleness was obtained by correlating the plastic strain with E and other relevant parameters, such as B19, which is essential for optimizing completion and stimulation strategies. These correlations enable us to identify areas of high brittleness, which can be targeted for specific evaluation and optimization, including determining the optimal landing point and selecting the most suitable interval for hydraulic fracturing stages and clusters to enhance production and reduce the risk of poor well performance. Furthermore, this integrated approach allows us to prioritize specific treatments, such as frac design and execution, to mitigate potential risks and maximize production outcomes, ultimately leading to more efficient and effective field development.

The analysis was complemented by incorporating the mineralogical composition of the rock into the brittleness-plasticity correlation. A detailed mineralogical assessment was performed for each sample to quantify the proportions of quartz, clay, and TOC. Understanding the mineralogical makeup is essential for evaluating mechanical behavior, as different minerals exhibit distinct elastic and plastic characteristics that directly influence rock brittleness. Quartz, a hard and rigid mineral with high E and low deformability, contributes significantly to the brittle nature of the rock. The brittleness-plasticity correlation was further elucidated by incorporating XRD data, which showed a strong correlation between brittleness and the mineralogical composition of the rock. Specifically, the results indicated that increasing quartz content and decreasing clay composition were associated with enhanced brittleness, highlighting the importance of mineralogical factors in controlling rock mechanical behavior, [Figure 8](#) and [Figure 9](#).

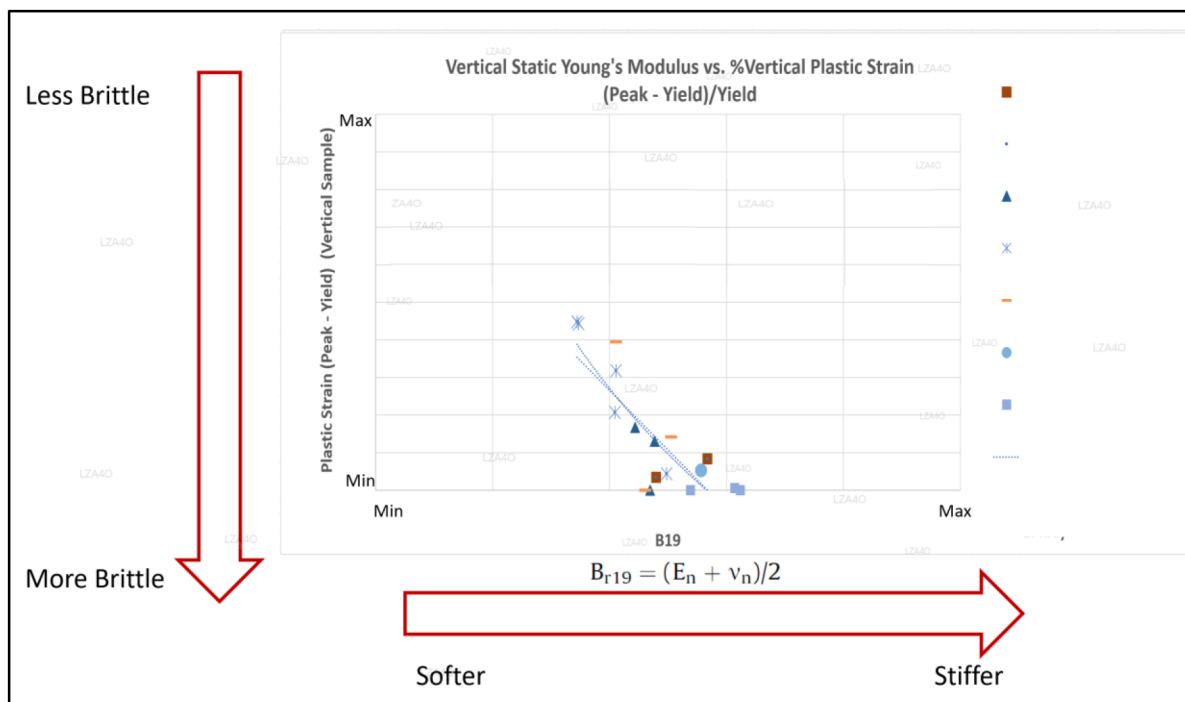


Figure 7— E vs plastic strain (inverse of brittleness).

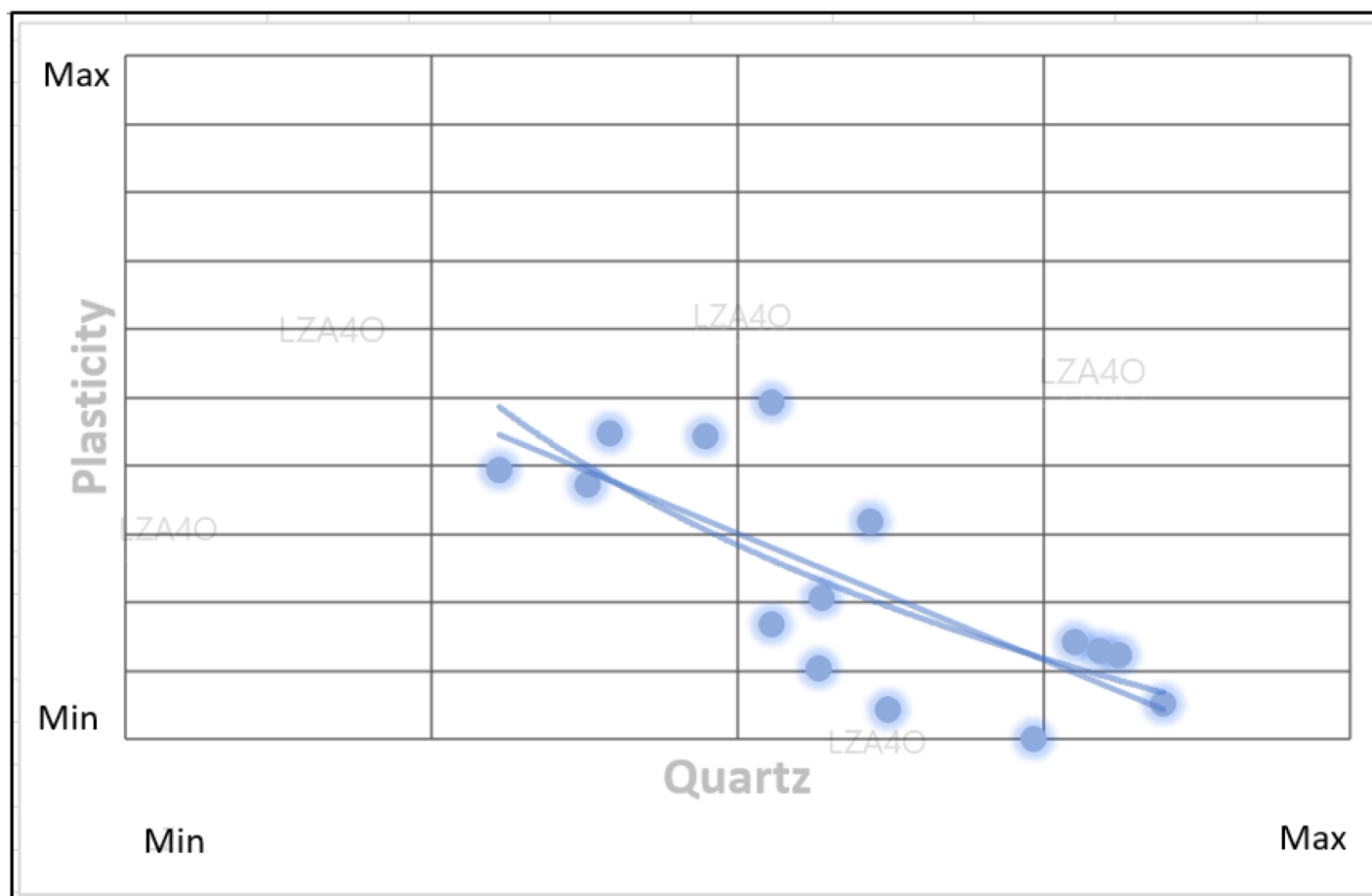


Figure 8—Quartz vs plastic strain (inverse of brittleness).

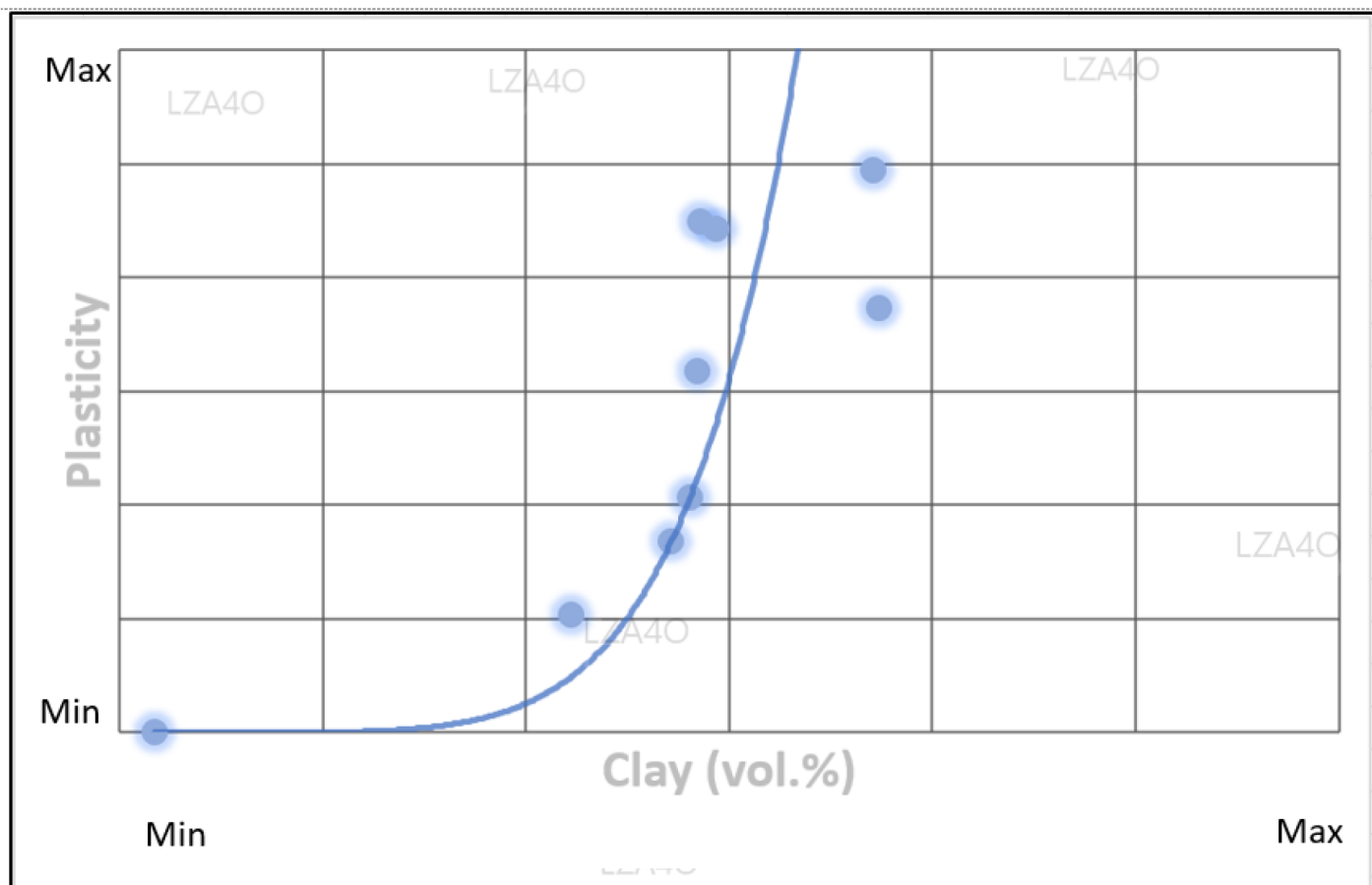


Figure 9—Clay vs plastic strain (inverse of brittleness).

In contrast, clay minerals are less brittle and to an extreme could be soft and ductile, and their presence tends to reduce brittleness by enhancing the rock's ability to deform plastically, [Figure 9](#).

The TOC content, representing the amount of organic matter within the rock, also plays a crucial role, as organic matter is generally softer and more compliant than the surrounding mineral matrix, thereby influencing both elasticity and plasticity, [Figure 10](#).

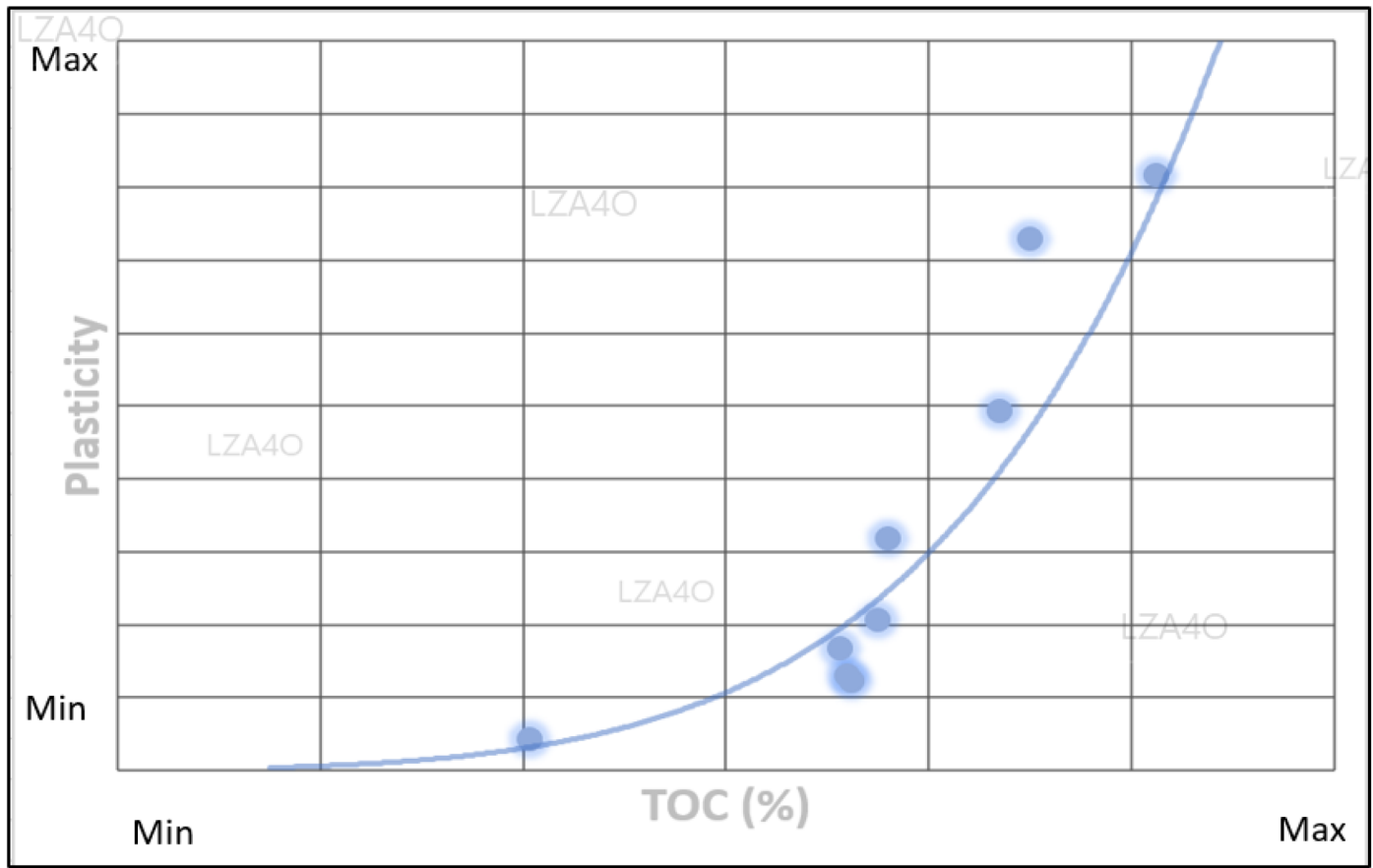


Figure 10—TOC vs plastic strain (brittleness).

A multivariable regression was developed to quantitatively capture the complex interplay between mineralogy and mechanical properties for estimating rock brittleness. This empirical equation integrates both mineralogical and geomechanical parameters, and is expressed as:

$$Plasticity = a * \ln(E) + b * \ln(B_{19}) + c * Q + d * eC + e^{TOC} + f \quad (5)$$

where Plasticity is the percentage of ductility (inverse of brittleness) Q is the quartz content, C is the clay content and f is a constant. Track 3 in Figure 11 shows a comparison of the computed multivariable brittleness vs. the measured brittleness in sample during triaxial test.

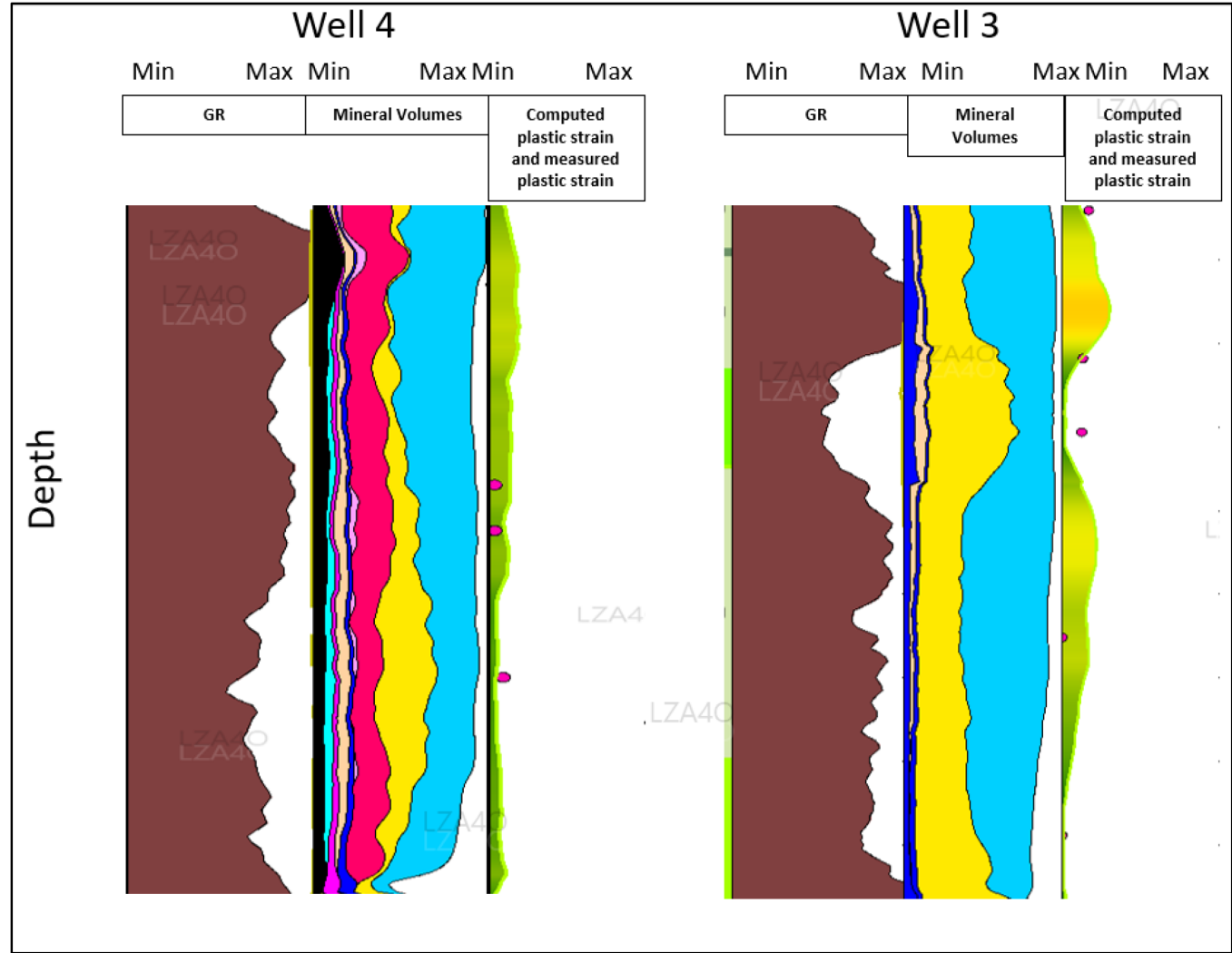


Figure 11—Track 1 (GR), track 2 (mineralogy) and track 3 (computed plastic strain (curves) with measured plastic strain (red disks) – inverse of brittleness).

Finally, a comprehensive analysis of each well's performance was conducted by normalizing the flowing head pressure and the gas rate by the choke size and the number of hydraulic fracturing stages. This allowed us to establish a standardized metric for evaluating well performance, which was then compared to the estimated brittleness of each well. The results of this analysis revealed a clear and significant correlation between well performance and brittleness, with wells exhibiting higher performance being preferentially located in the most brittle rock formations, [Figure 12](#) and [Figure 13](#).

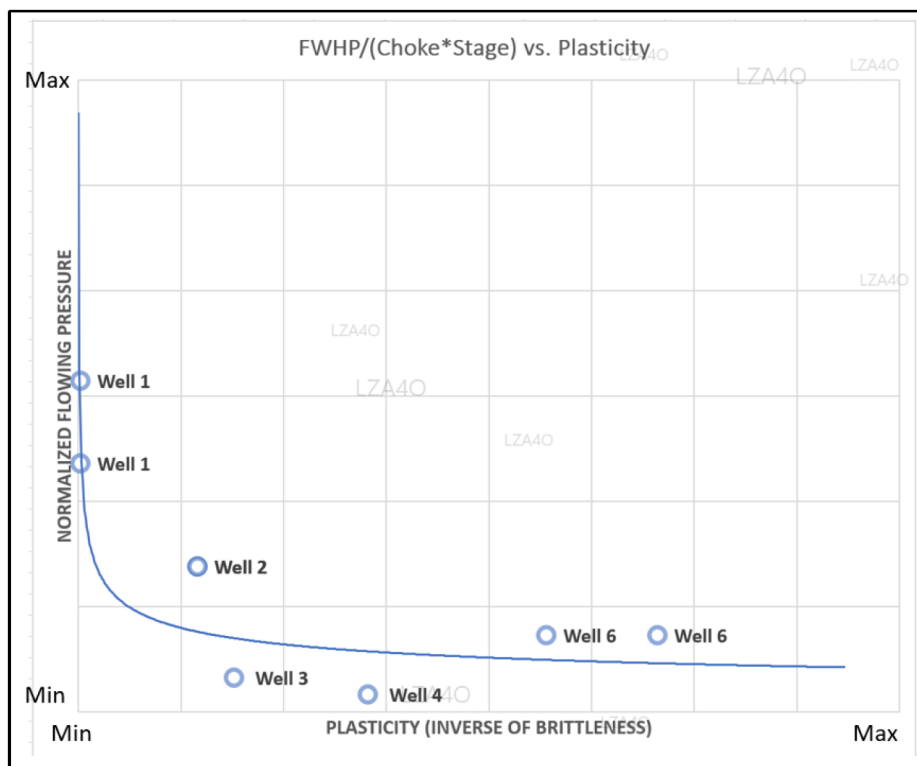


Figure 12—Brittleness vs normalized flowing pressure (well head pressure normalized by choke size and the number of hydraulic fracturing stages).

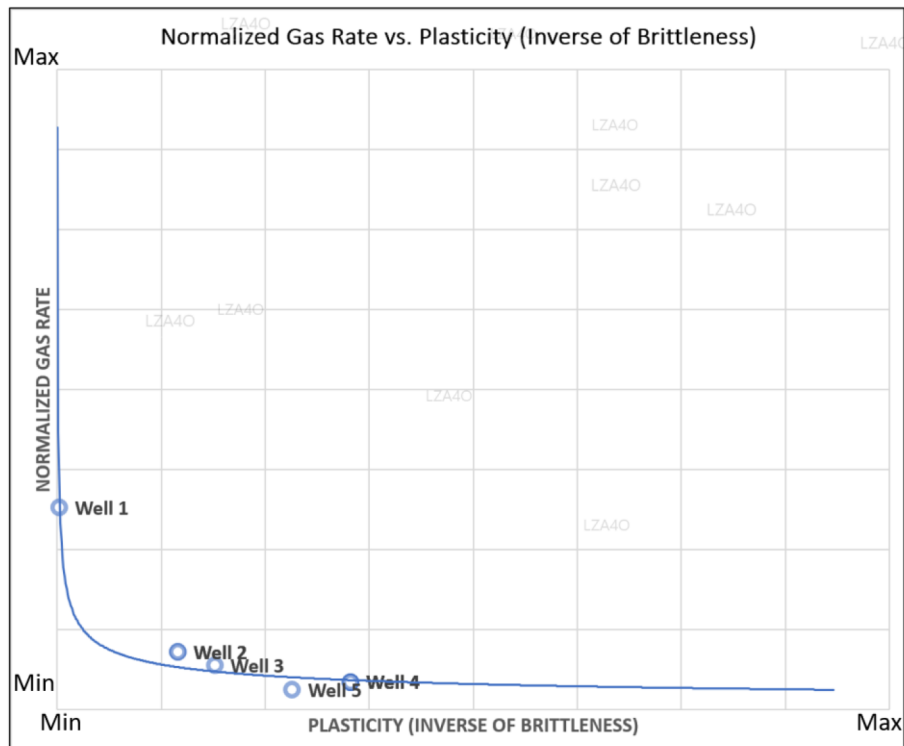


Figure 13—Brittleness vs normalized gas rate (gas rate normalized by choke size and the number of hydraulic fracturing stages).

These correlations suggest that the brittle nature of the rock plays a crucial role in determining the effectiveness of hydraulic fracturing and the subsequent well performance. The correlation between well

flowing pressure and gas rate with brittleness can be attributed to the fact that brittle rocks are more prone to fracturing and are more likely to create complex fracture networks. This, in turn, can lead to higher well performance, as the fractures provide a pathway for fluid flow and increase the rock's permeability. On the other hand, ductile rocks are less prone to fracturing and are more likely to deform plastically, which can lead to lower well performance. Specifically, the data indicate that wells placed in more brittle rock tend to have higher flowing head pressures and overall better production performance, likely due to the increased complexity and conductivity of the fracture networks created in these formations. Conversely, wells placed in less brittle rock tend to have lower performance, potentially due to the reduced effectiveness of hydraulic fracturing in these formations. These findings have important implications for optimizing well placement, completion design, and hydraulic fracturing strategies to maximize production outcomes and minimize costs. This highlights the importance of accurate brittleness characterization in optimizing hydraulic fracturing operations and maximizing recovery from unconventional reservoirs.

Conclusions

The results of this study have significant implications for the oil and gas industry, as they highlight the importance of accurate brittleness characterization in optimizing hydraulic fracturing operations. By using a combination of rock mechanics testing, mineralogical analysis, and empirical correlations, operators can gain a better understanding of the brittleness of their reservoir rocks and optimize their hydraulic fracturing operations accordingly. This can lead to improved well performance, increased recovery, and reduced costs. In conclusion, the concept of brittleness is a critical parameter in the oil and gas industry, particularly in unconventional reservoirs. By using a combination of rock mechanics testing, mineralogical analysis, and empirical correlations, operators can gain a better understanding of the brittleness of their reservoir rocks and optimize their hydraulic fracturing operations accordingly. The results of this study highlight the importance of accurate brittleness characterization in optimizing hydraulic fracturing operations and maximizing recovery from unconventional reservoirs.

The study also highlights the importance of considering the mineralogical composition of the rock in estimating brittleness. The mineralogical composition of the rock can have a significant impact on its brittleness, and ignoring this can lead to inaccurate estimates of brittleness. By considering the mineralogical composition of the rock, operators can gain a more accurate understanding of the brittleness of their reservoir rocks and optimize their hydraulic fracturing operations accordingly.

Overall, the results of this study provide a comprehensive understanding of the concept of brittleness and its importance in the oil and gas industry. By considering the mineralogical composition of the rock, using a combination of rock mechanics testing and empirical correlations, and optimizing hydraulic fracturing operations based on accurate estimates of brittleness, operators can maximize recovery from unconventional reservoirs and improve well performance.

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